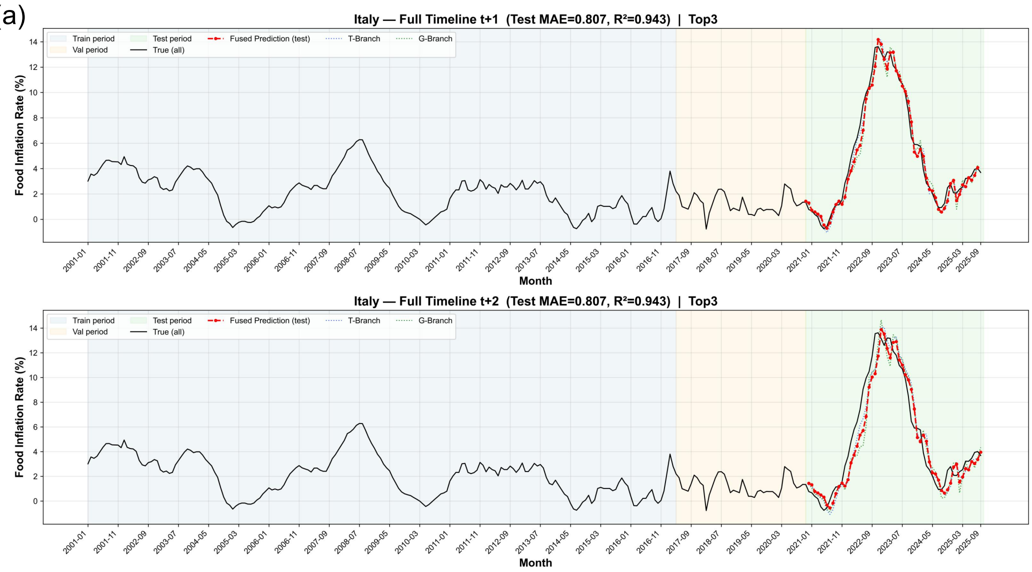
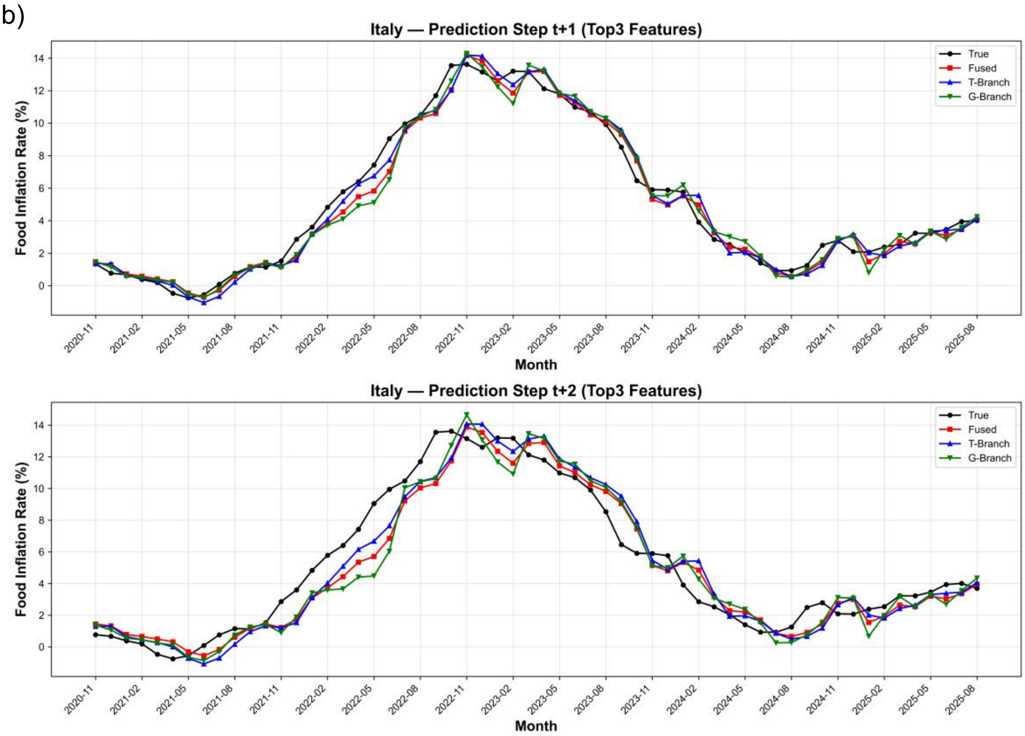
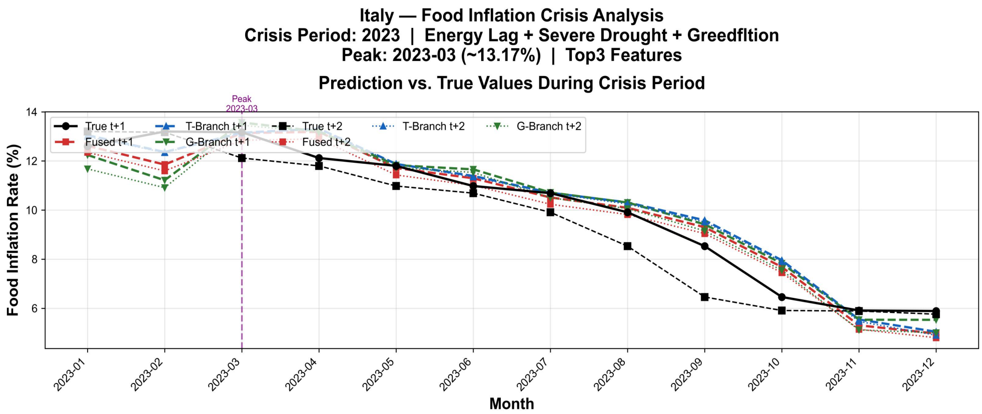
(a)


Italy — Full Timeline t+1 (Test MAE=0.807, R²=0.943) | Top3
Train period
Val period
Test period
True (all)
Fused Prediction (test)
T-Branch
G-Branch
Food Inflation Rate (%)
Month
Italy — Full Timeline t+2 (Test MAE=0.807, R²=0.943) | Top3
Train period
Val period
Test period
True (all)
Fused Prediction (test)
T-Branch
G-Branch
Food Inflation Rate (%)
Month

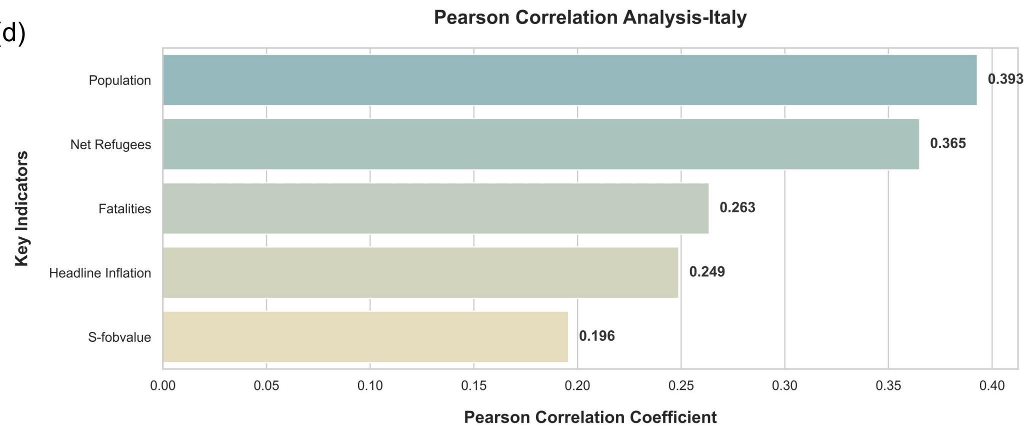
(b)


Italy — Prediction Step t+1 (Top3 Features)
True
Fused
T-Branch
G-Branch
Food Inflation Rate (%)
Month
Italy — Prediction Step t+2 (Top3 Features)
True
Fused
T-Branch
G-Branch
Food Inflation Rate (%)
Month

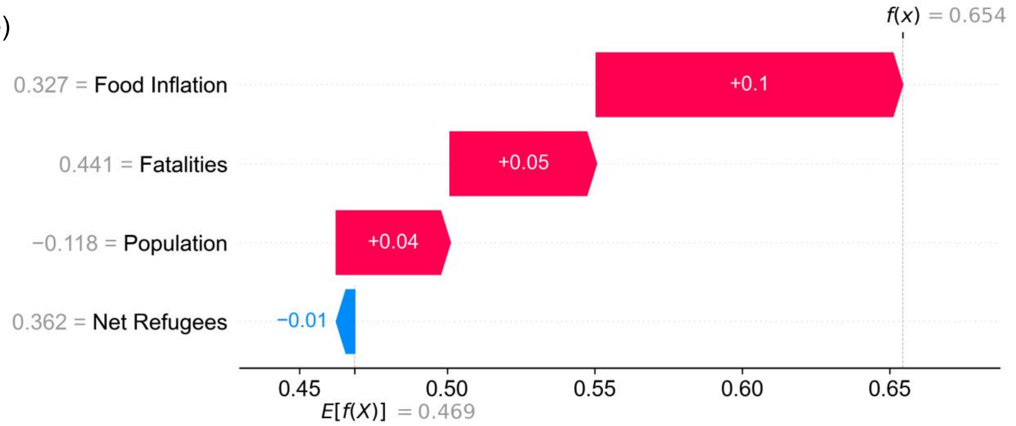
(c)


Italy — Food Inflation Crisis Analysis
Crisis Period: 2023 | Energy Lag + Severe Drought + Greedfltion
Peak: 2023-03 (~13.17%) | Top3 Features
Prediction vs. True Values During Crisis Period
Peak
2023-03
True t+1
Fused t+1
T-Branch t+1
G-Branch t+1
True t+2
Fused t+2
T-Branch t+2
G-Branch t+2
Food Inflation Rate (%)
Month

(d)



(e)



Multidimensional Evaluation and Attribution Analysis of Food Inflation Forecast in Italy. (a) Overview of the total cycle prediction and fitting, (b) Multi-step Short-Term Forecast Comparison Chart, (c) Details of Prediction Performance in Extreme Crisis Periods, (d) Pearson correlation analysis of macroeconomic drivers, (e) SHAP-based explanation of feature contributions for a specific prediction instance.

Frequent extreme heat and prolonged droughts in recent years have decimated Italy's iconic durum wheat, olive, and wine grape yields, driving up raw material costs for staples such as pasta and olive oil. As one of the European countries with the highest energy costs, the Italian food processing industry faces substantial electricity expenditures for drying, packaging, and refrigeration, with energy price fluctuations reflected on shelf labels in near real time. Furthermore, the Italian food system is characterized by a high proportion of small and medium-sized enterprises (SMEs), which possess limited resilience against raw material volatility and rising financing costs, often forcing a rapid pass-through of cost pressures to consumers. This synergy between climate-induced commodity shortages and an energy-dependent processing structure has forged a unique inflation pattern, defined by the price leadership of specialty agricultural products.